Conceptual Design of Aircraft
Propeller – Further examples

School of Engineering & Materials Science, Queen Mary University of London

1. A propeller is known to have the following properties: Four blades, diameter of 2 m, frequency of n = 20 Hz, activity factor of 160, mean drag and integrated design lift coefficients of 0.1 and 0.3 respectively. Calculate:

a) Static power and torque coefficients

b) Static thrust coefficient

c) Static thrust at standard sea level conditions

2. The same propeller flies at a speed of 50 m/s and in standard sea level conditions. The air velocity downstream of the propeller is 52.5 m/s. Calculate:

a) The advance ratio J

b) The thrust

c) The thrust coefficient

Standard sea level conditions:

$\rho = 1.225$ kg/m^3, p = 101.325 kPa, T = 288.16 K, c = 340.3 m/s

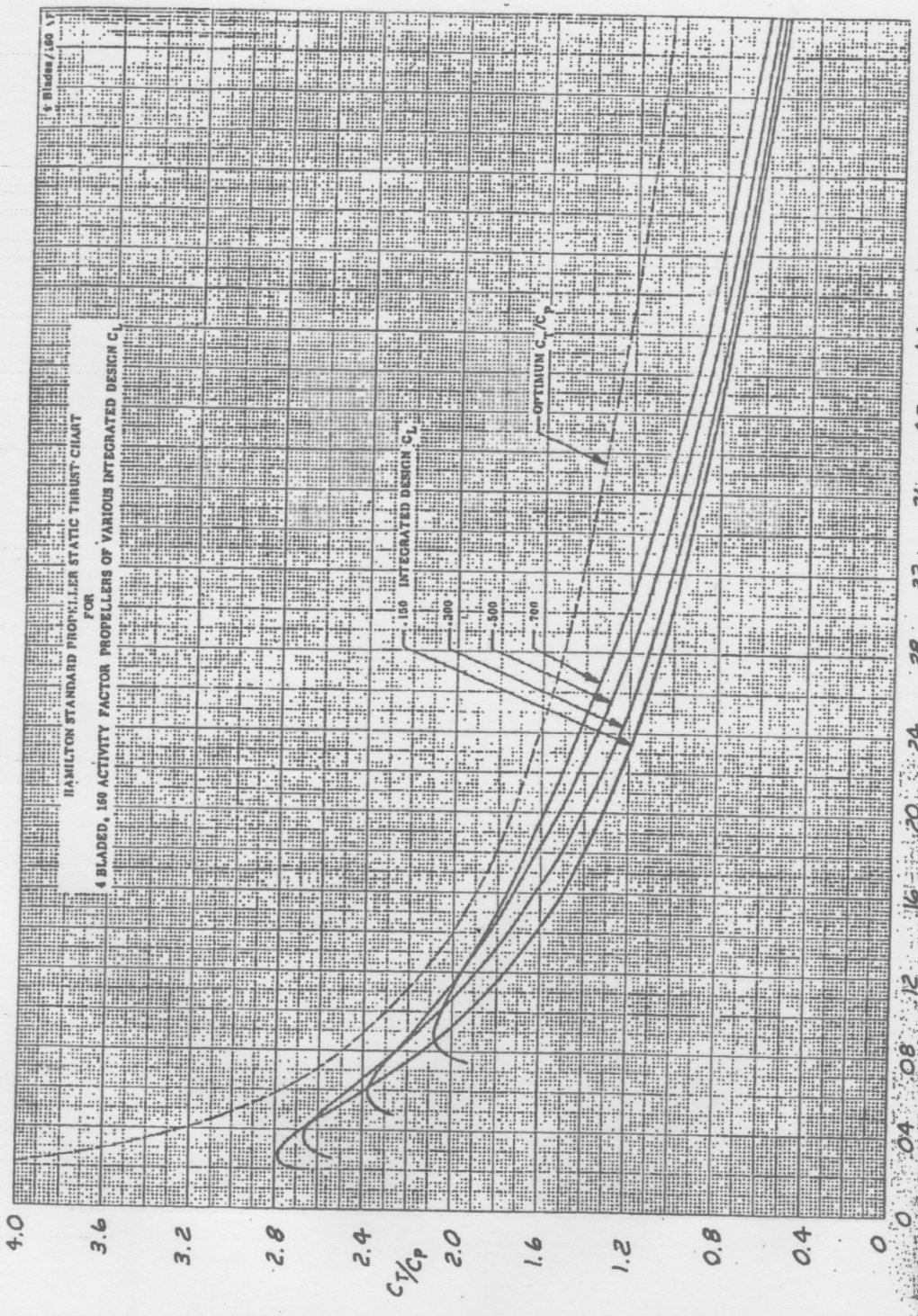


Chart C7 **Propeller Thrust and Power Coefficients for Static Thrust with B=4 and AF=160**

Eldad Avital

1.

Data:

B = 4
D = 2 m
n = 20 Hz
AF = 160
$\bar{C}_d$ = 0.1
C_{Li} = 0.3

Find:

a) Static power and torque coefficients, C_P and C_Q
b) Static thrust coefficient, C_T
c) Static thrust F at standard sea level conditions

Solution:

Power coefficient

$$P = 4\,B\,\rho\,\pi^3 n^3 \bar{C}_d \left(\frac{D}{10}\right)^5 AF$$

$$C_P = \frac{P}{\rho\, n^3 D^5}$$

$$C_P = 4\,B\,\pi^3 \bar{C}_d\, AF \cdot 10^{-5}$$

$$C_P = 4 \cdot 4 \cdot \pi^3 \cdot 0.1 \cdot 160 \cdot 10^{-5} = 0.08$$

Torque coefficient

$$Q = \frac{P}{2\,\pi\, n}$$

$$C_Q = \frac{Q}{\rho\, n^2 D^5}$$

$$C_Q = \frac{1}{2\,\pi}\frac{P}{\rho\, n^3 D^5} = \frac{C_P}{2\,\pi}$$

$$C_Q = \frac{0.08}{2\,\pi} = 0.01263$$

Static thrust coefficient

From the static thrust and power coefficient chart, for C_P = 0.08 and C_{Li} = 0.3:

$$\frac{C_T}{C_P} = 2.25$$

$$C_T = 2.25 \cdot 0.08 = 0.18$$

$$F = C_T\, \rho\, n^2 D^4$$

$$F = 0.18 \cdot 1.225 \cdot 20^2 \cdot 2^4$$

$$F = 1411.2 \quad N$$

2.

Data:

Va = 50 m/s

Vout = 52.5 m/s

Find: J, F and C_T

Solution:

$$J=\frac{V_a}{n\,D}=\frac{50}{20\cdot 2}=1.25$$

$$\eta_p=\frac{2V_a}{V_a+V_{out}}$$

$$\eta_p=\frac{2\cdot 50}{50+52.5}=0.9756$$

$$\eta_p=\frac{F\,V_a}{P}$$

Assume P ≠ f(Va), therefore:

$$P=C_P\,\rho\,n^3\,D^5$$

$$0.9756=\frac{F\cdot 50}{0.08\cdot 1.225\cdot 20^3\cdot 2^5}$$

$$F=489.8\quad N$$

$$C_T=\frac{F}{\rho\,n^2\,D^4}$$

$$C_T=\frac{489.8}{1.225\cdot 20^2\cdot 2^4}$$

$$C_T=0.0625$$